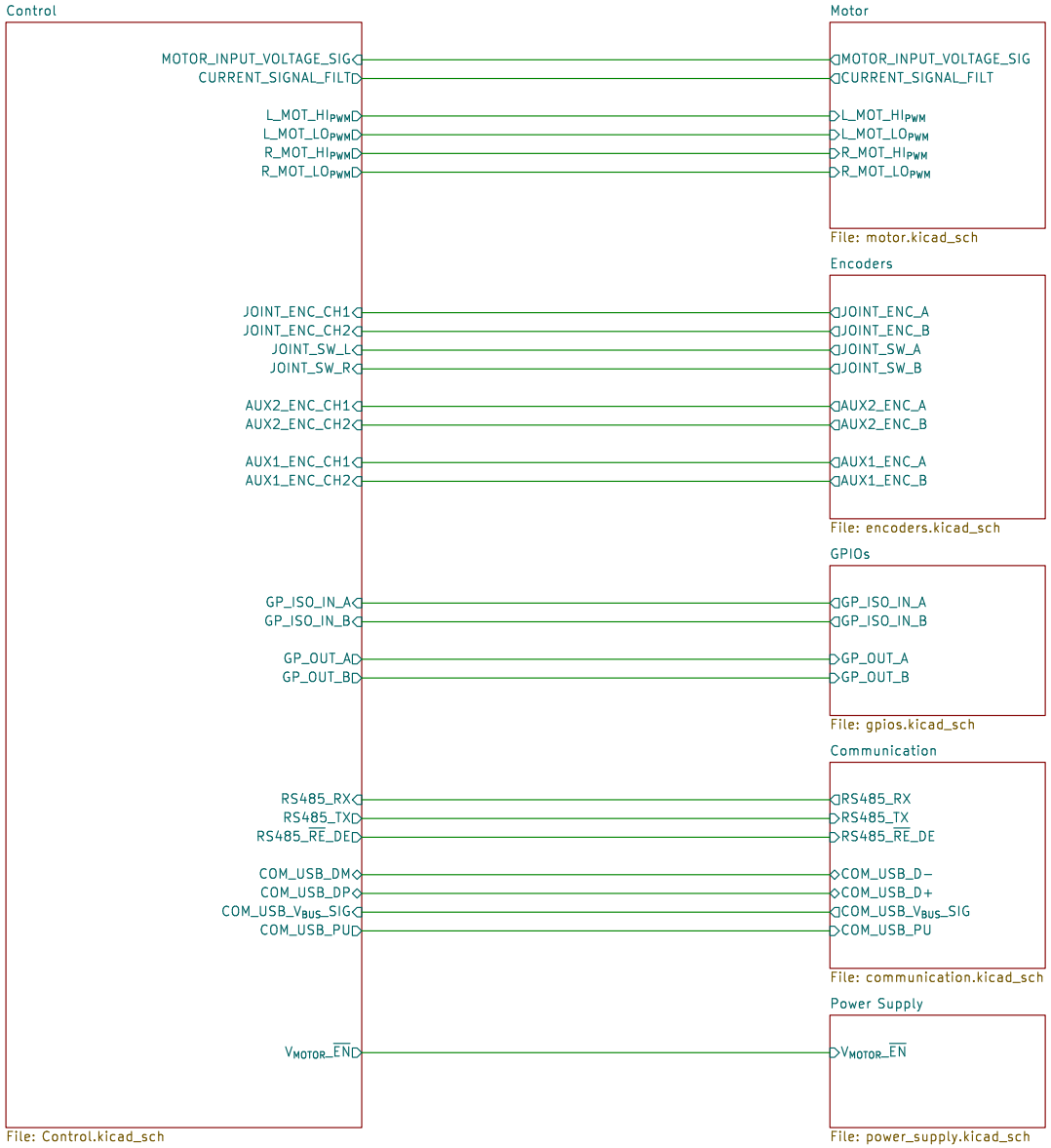
System overview



H1 H3
H2 H4

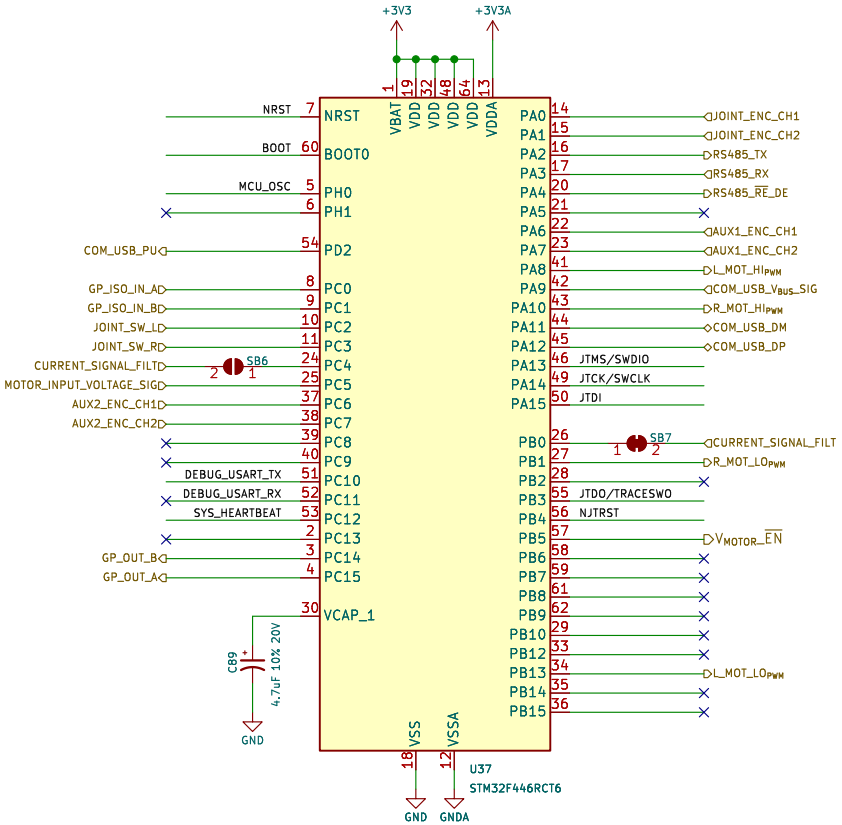
Single Joint Controller Board
Designed by Alves, T. G.

Sheet: /
File: hardware.kicad_sch

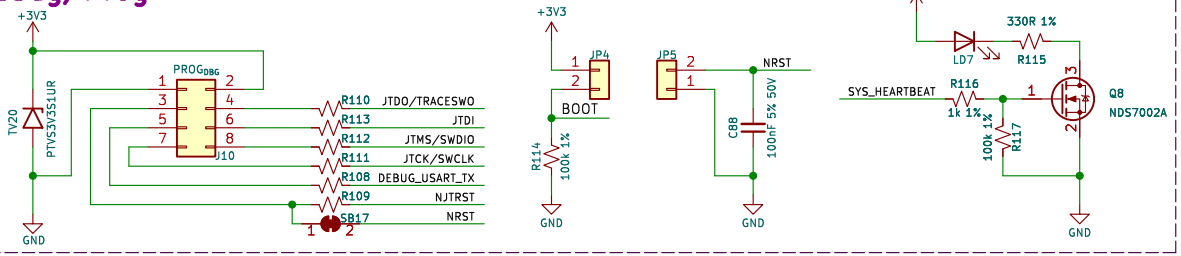
Title: System overview

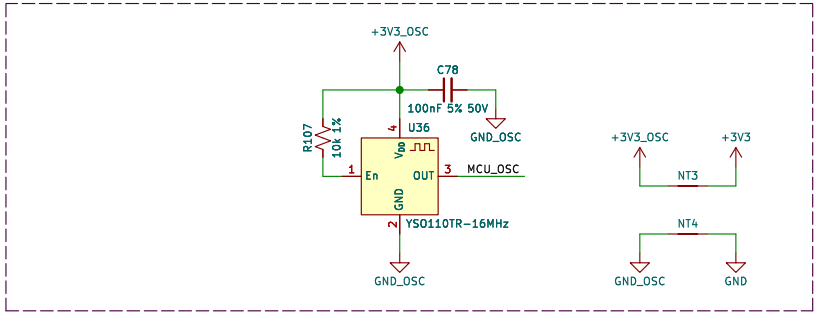
Size: A4	Date: 2023-05-30	**Rev: 0.1**
KiCad E.D.A. kicad (7.0.0)		Id: 1/7

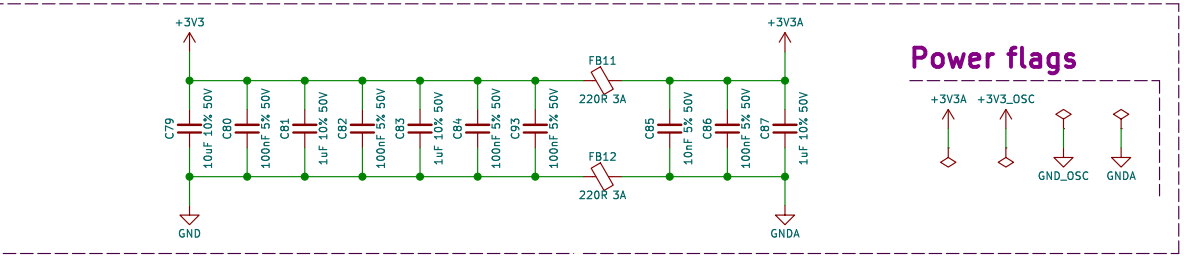
Microcontroller



Debug/Prog







Power flags

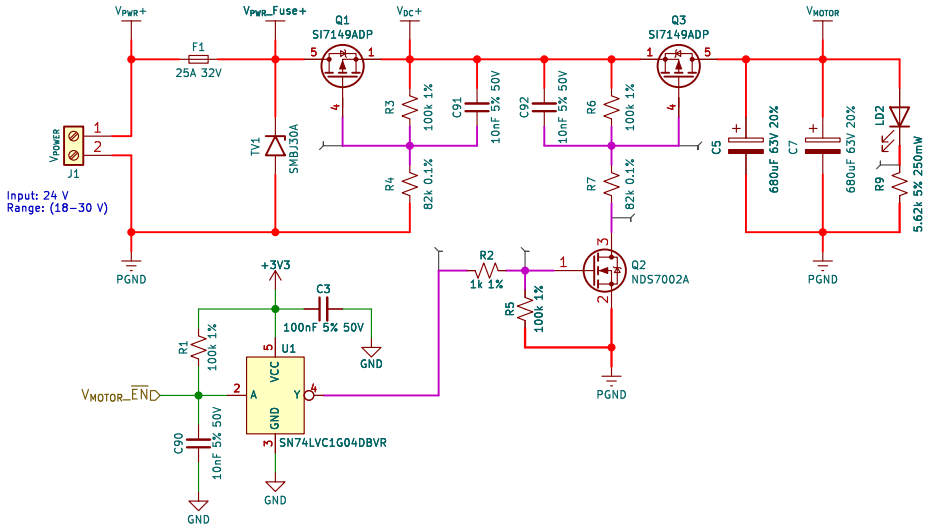
Single Joint Controller Board
Designed by Alves, T. G.

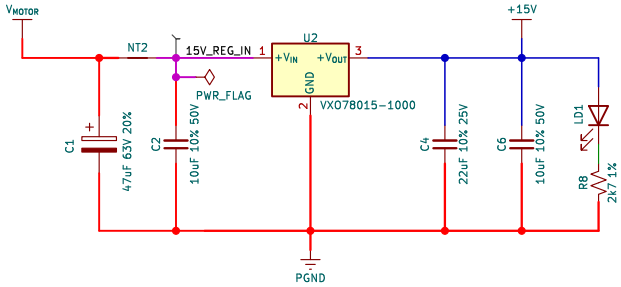
Sheet: /Control/
File: Control.kicad_sch

Title: SJC Control

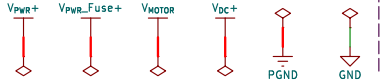
Size: A4	Date: 2023-05-30	**Rev: 0.1**
KiCad E.D.A. kicad (7.0.0)		Id: 2/7

Power Inputs

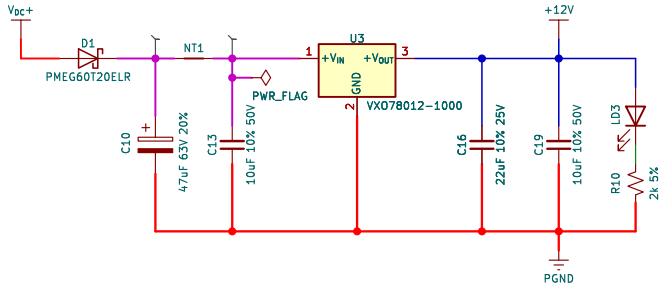


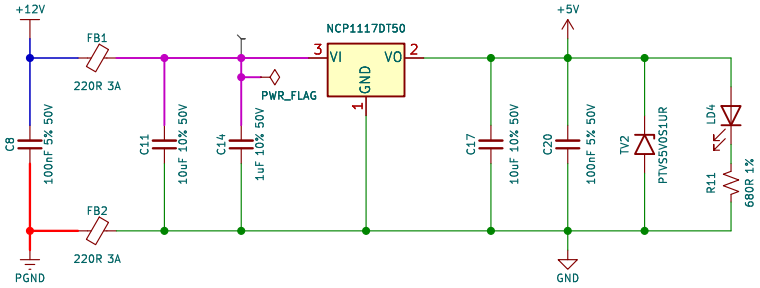


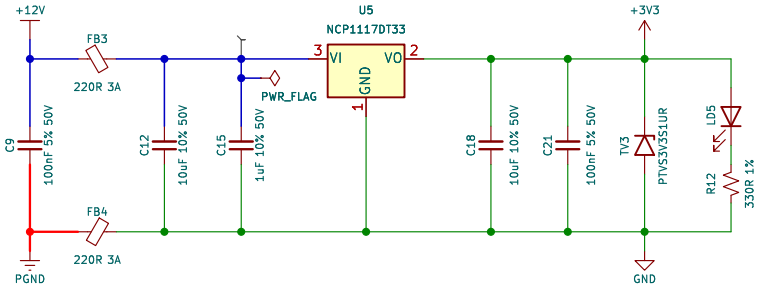
Power flags



Power Inputs







Single Joint Controller Board
Designed by Alves, T. G.

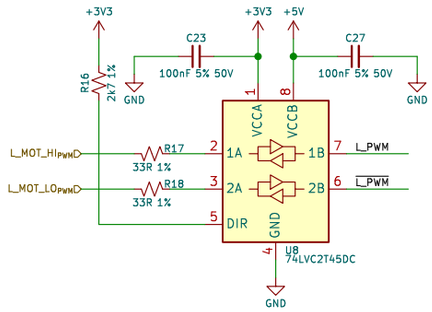
Sheet: /Power Supply/
File: power_supply.kicad_sch

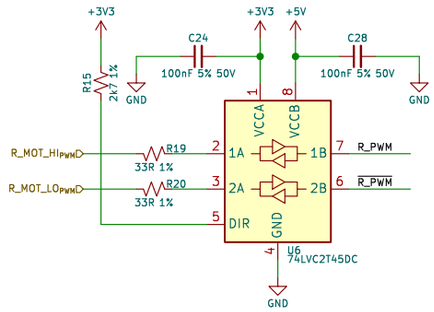
Title: SJC Power Supply

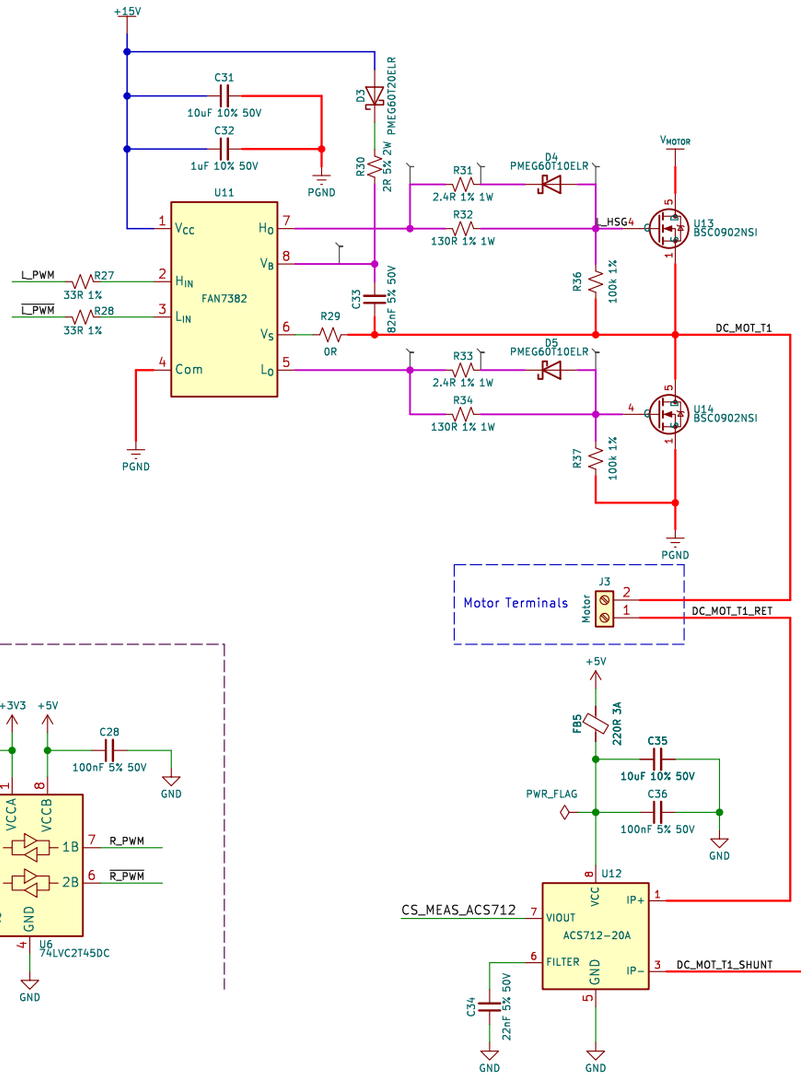
Size: A4	Date: 2023-05-30	**Rev: 0.1**
KiCad E.D.A. kicad (7.0.0)		Id: 3/7

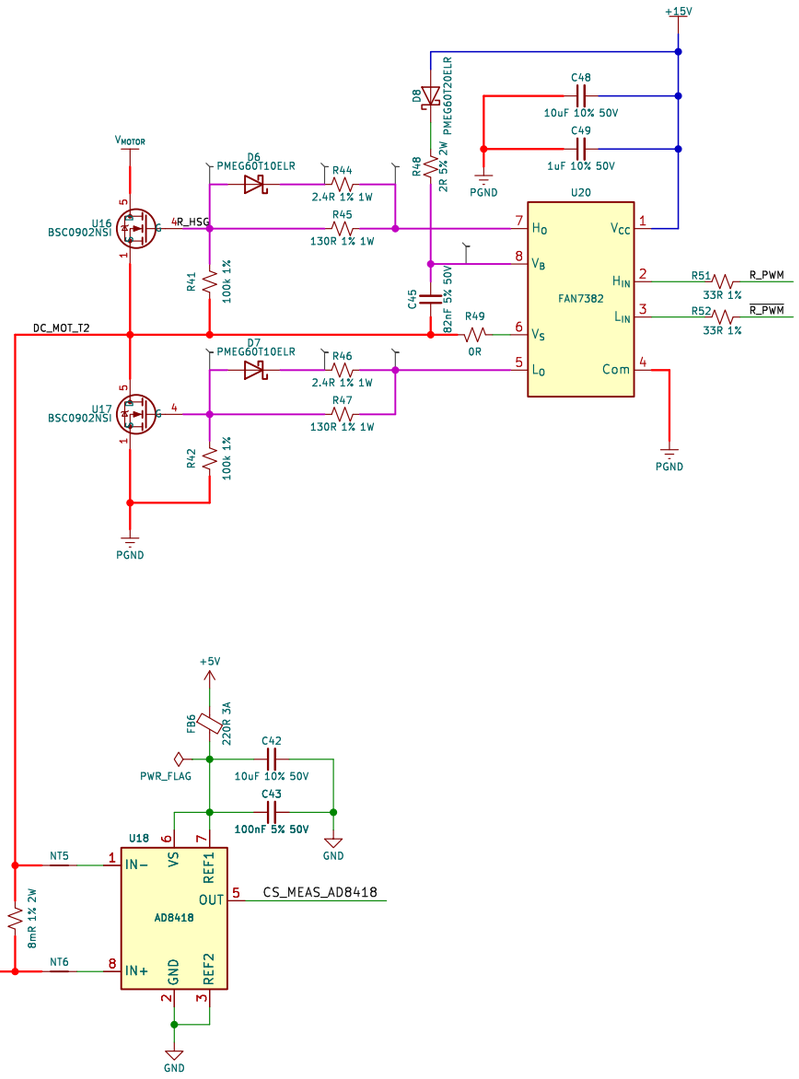
Servomotor interface

PWM Signals

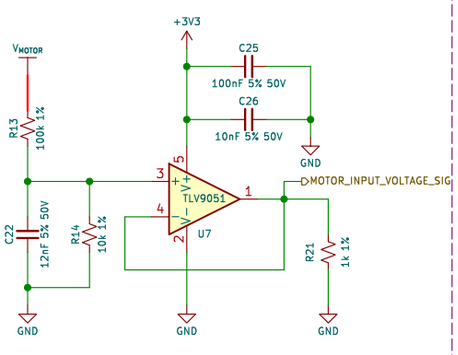






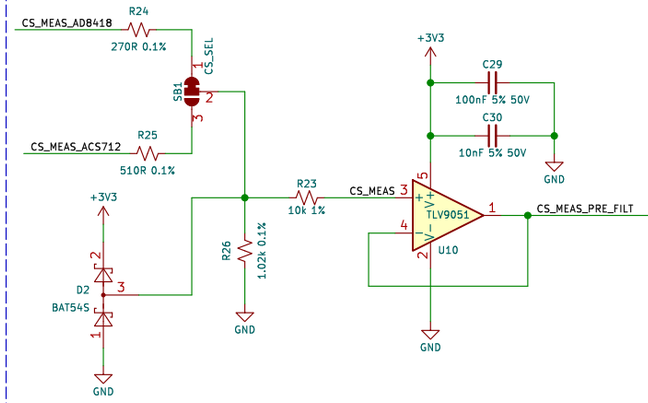


H-Bridge input voltage sense

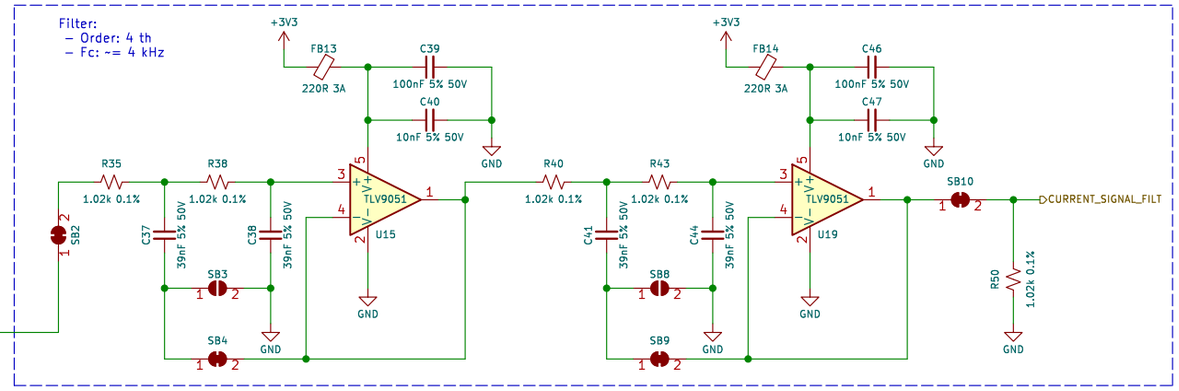


Current signal conditioning interface

Voltage conditioner:
- AD8418: Av = 0.80 V/V
- ACS712: Av = 0.67 V/V



Filter:
- Order: 4 th
- Fc: ~= 4 kHz



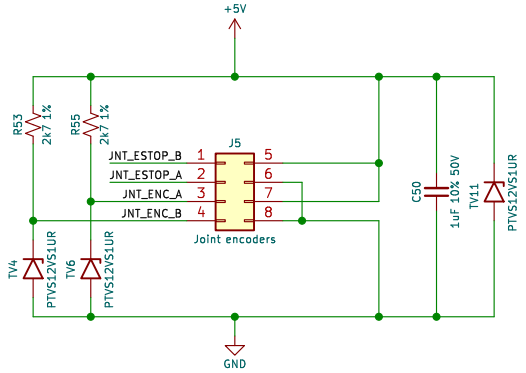
Single Joint Controller Board
Designed by Alves, T. G.

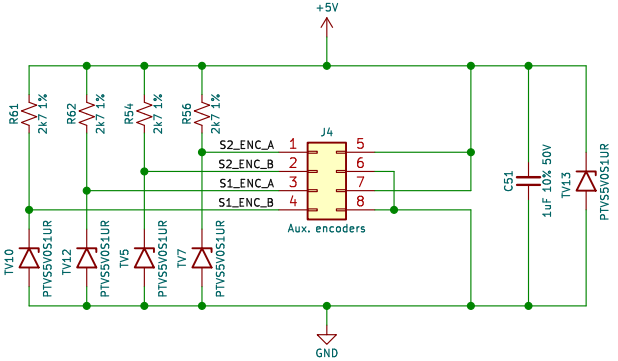
Sheet: /Motor/
File: motor.kicad_sch

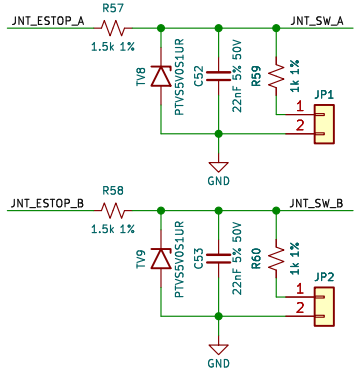
Title: SJC Servomotor

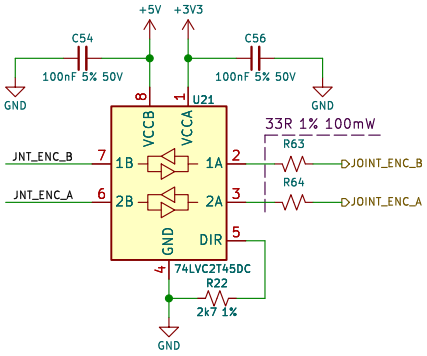
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KiCad E.D.A. kicad (7.0.0)		Id: 4/7

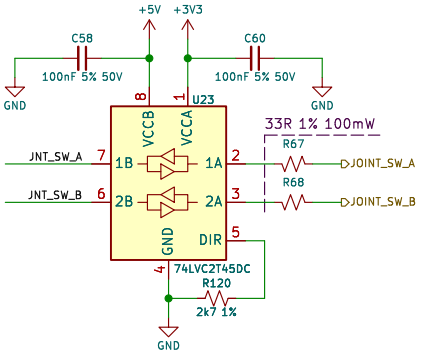
Encoder interfaces

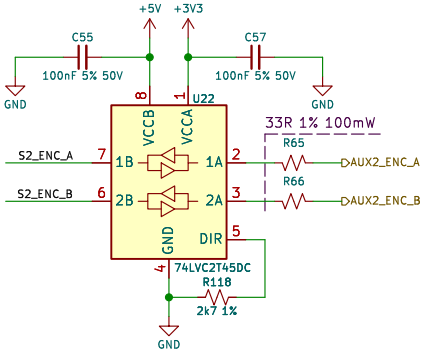


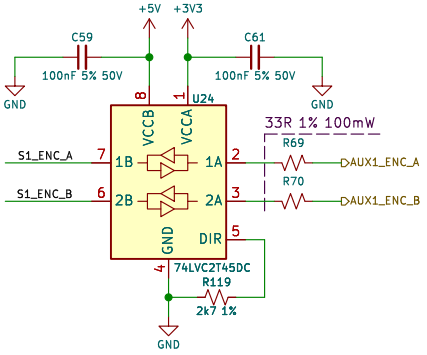












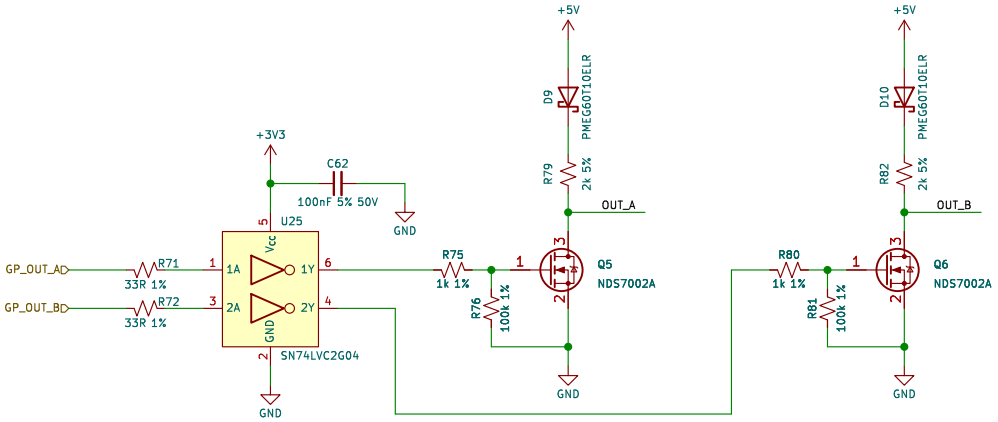
Single Joint Controller Board
Designed by Alves, T. G.

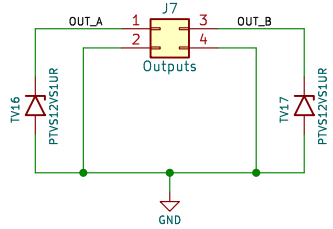
Sheet: /Encoders/
File: encoders.kicad_sch

Title: SJC Encoders

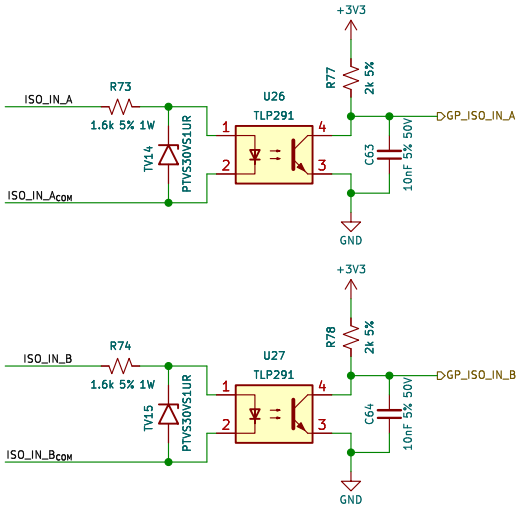
Size: A4	Date: 2023-05-30	**Rev: 0.1**
KiCad E.D.A. kicad (7.0.0)		Id: 5/7

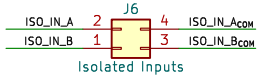
Digital Outputs





Isolated Inputs





Single Joint Controller Board
Designed by Alves, T. G.

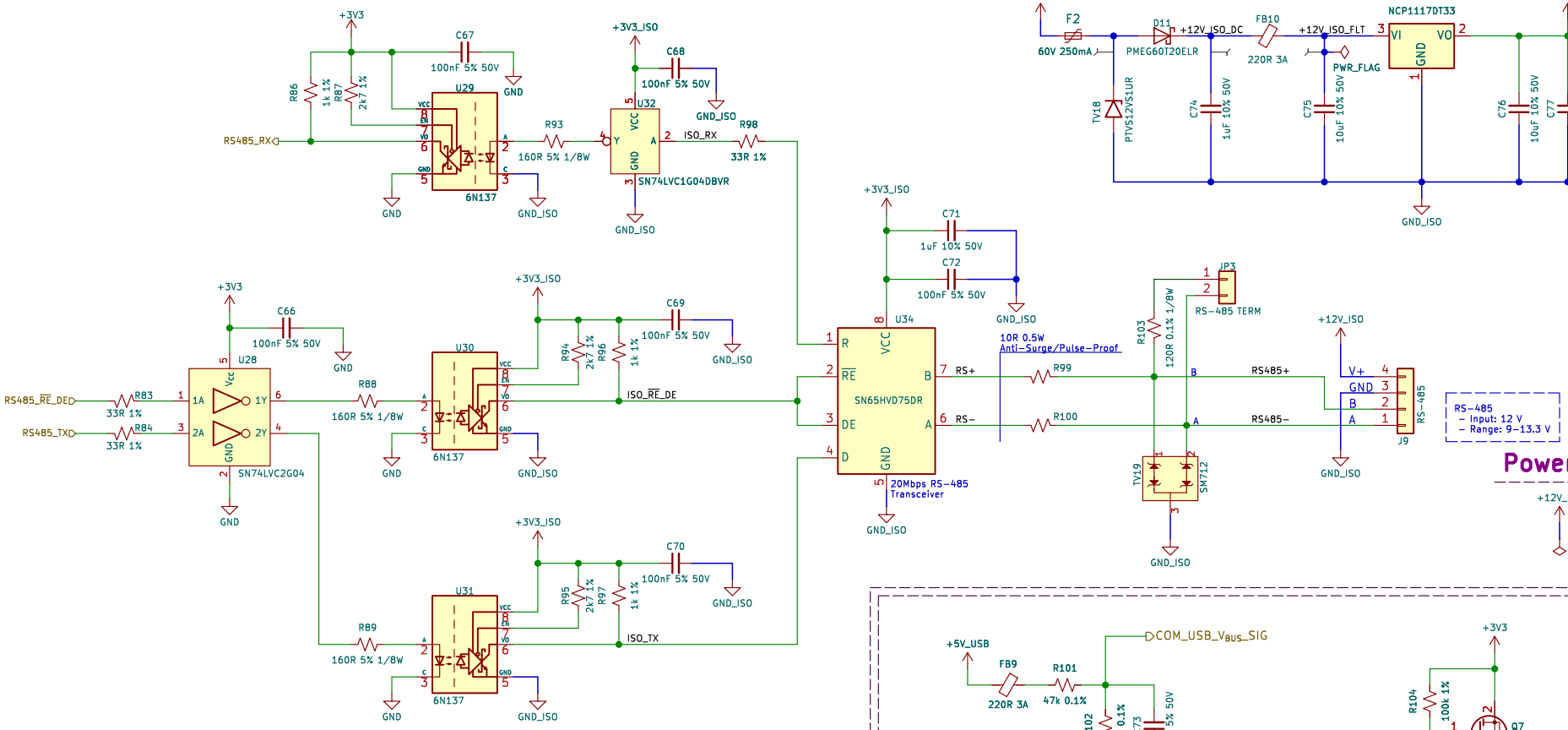
Sheet: /GPIOs/
File: gpios.kicad_sch

Title: SJC I/Os

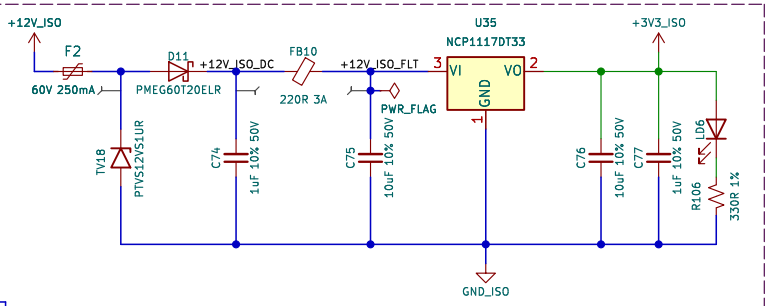
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KiCad E.D.A. kicad (7.0.0)		Id: 6/7

Communication

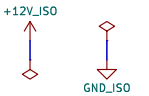
RS-485 (ISO)



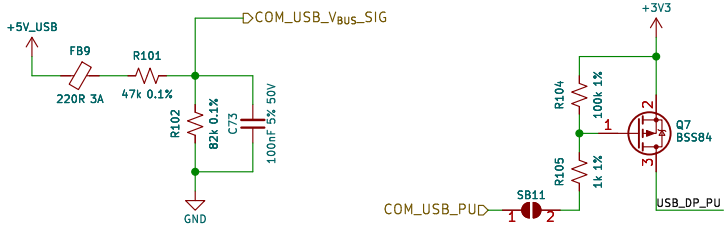
BUS power supply



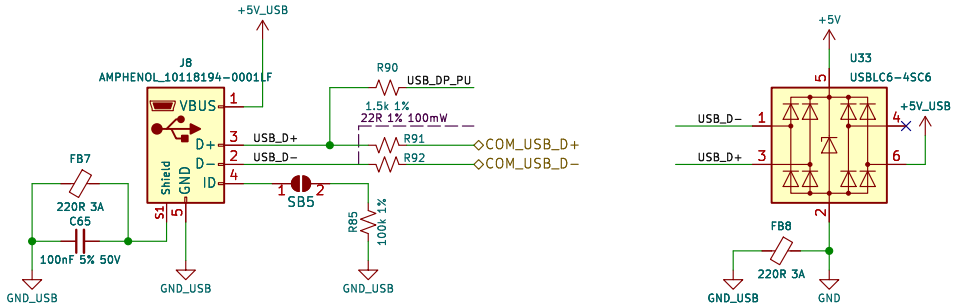
Power flags



RS-485
- Input: 12 V
- Range: 9-13.3 V



USB



Single Joint Controller Board
Designed by Alves, T. G.

Sheet: /Communication/
File: communication.kicad_sch

Title: SJC Communication

Size: A4	Date: 2023-05-30	**Rev: 0.1**
KiCad E.D.A. kicad (7.0.0)		Id: 7/7